ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 11

AGRICULTURAL TECHNOLOGY

MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 11 Agricultural Technology practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 11. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Penalise once per question for missing units. Accept other correct answers.

QUESTION 1

1.1 1.1.1 A ✓✓ 1.1.2 A ✓✓ 1.1.3 C ✓✓ 1.1.4 B ✓✓ 1.1.5 A ✓✓ (10)

1.2 1.2.1 Hydraulics (hydraulic system) ✓✓ 1.2.2 Head (lift height) ✓✓ 1.2.3 Thermoplastic ✓✓ 1.2.4 Preventive (scheduled) maintenance ✓✓ 1.2.5 Transformer ✓✓ (10)

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QUESTION 2

2.1 Thermoplastics soften when heated and can be reshaped many times ✓, e.g. polythene, PVC, nylon ✓. Thermosetting plastics set permanently when first heated and cannot be remelted ✓, e.g. Bakelite, epoxy resin, melamine ✓. (4)

2.2 Any THREE: irrigation pipes ✓; tunnel / greenhouse covers ✓; water tanks ✓; silage wrap; mulch sheeting; feed bags; ear tags (3)

2.3 MIG, any ONE: fast, continuous welding; little slag to clean; easy to learn; good for thin metal ✓✓. Arc, any ONE: cheap and portable equipment; works outdoors in wind; suitable for thick and dirty metal ✓✓. (4)

2.4 Any TWO with a cause: porosity (gas holes) ✓ → dirty metal or damp electrodes ✓; undercut ✓ → current too high or wrong angle ✓; slag inclusion → slag not cleaned between runs; cracks → cooling too fast; lack of fusion → current too low or welding too fast (4)

2.5 Cast iron is brittle ✓. Preheating reduces the temperature difference between the weld and the rest of the metal ✓, so it cools slowly and evenly and does not crack ✓. (3)

2.6 Any TWO: visual inspection ✓; dye penetrant test ✓; X-ray (radiographic) test; ultrasonic test; magnetic particle test (2)

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QUESTION 3

3.1 Any FOUR: engine oil level ✓; coolant (radiator) level ✓; tyre pressure and condition ✓; fuel level ✓; hydraulic oil level; air filter; brakes and lights; leaks (4)

3.2 Pressure applied to a liquid in a closed system is passed on equally in all directions ✓✓. The hydraulic pump pushes oil into a cylinder; the pressure acts on a large piston and creates a large force ✓ that lifts the three-point linkage and the implement ✓. (4)

3.3 Any THREE: keep the PTO shaft guards in place ✓; switch off the PTO and engine before working on the implement ✓; no loose clothing near the shaft ✓; never step over a turning shaft; keep bystanders away (3)

3.4 (a) $45 \div 6$ ✓ $= 7{,}5$ days ✓ (5)
(b) $45 \times 12 = 540\ \ell$ ✓; $540 \times \text{R}23$ ✓ $= \text{R}12\ 420$ ✓

3.5 Correct pressure gives the best grip (traction) and less soil compaction ✓, reduces tyre wear and fuel use ✓. (2)

3.6 Any TWO: mouldboard plough ✓; disc plough ✓; chisel plough; subsoiler; ripper (2)

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QUESTION 4

4.1 Any FOUR: energiser ✓; earth system (earth spikes) ✓; insulators ✓; fence wires ✓; lightning diverter; warning signs; cut-out switches (4)

4.2 The circuit is completed through the soil and the earth system ✓; poor earthing gives a weak shock, so animals are not kept in ✓. (2)

4.3 (a) $\text{P} = \text{VI} = 230 \times 8{,}7$ ✓ $= 2\ 001\ \text{W}\ (\approx 2\ \text{kW})$ ✓ (6)
(b) $2{,}001\ \text{kW} \times 5\ \text{h}$ ✓ $= 10{,}01\ \text{kWh}$ ✓
(c) $10{,}01 \times \text{R}2{,}80$ ✓ $= \text{R}28{,}03$ ✓

4.4 Animal manure (and plant waste) is mixed with water in a sealed tank ✓. Bacteria break it down without oxygen (anaerobic digestion) ✓, producing gas that collects at the top ✓. Products: biogas (methane) for cooking, heating or electricity ✓; slurry used as fertiliser ✓. (5)

4.5 Any TWO: free energy from the sun ✓; works far from the electricity grid and during load shedding ✓; low maintenance; no fuel costs; no pollution (2)

4.6 It cuts off the power quickly when current leaks to earth, protecting people from electric shock ✓. (1)

[20]

QUESTION 5

5.1 $4 + 26$ ✓ $= 30\ \text{m}$ ✓ (2)

5.2 $\text{p} = 1\ 000 \times 9{,}8 \times 30$ ✓✓ $= 294\ 000\ \text{Pa} = 294\ \text{kPa}$ ✓ (3)

5.3 Centre pivot, any ONE: irrigates very large areas automatically with little labour ✓✓. Drip, any ONE: uses the least water / water goes straight to the roots / fewer weeds ✓✓. (4)

5.4 Too much water makes weak, porous concrete that cracks ✓✓; too little water makes concrete hard to work and it does not set properly ✓. (3)

5.5 Concrete is weak in tension ✓; the steel mesh strengthens it and prevents cracking when the trough is full of water ✓. (2)

5.6 Any THREE: a large catchment area with enough runoff ✓; a narrow valley for a short, cheap wall ✓; soil that holds water (clay) ✓; solid foundation for the wall; close to where the water is needed; does not flood important land (3)

5.7 A centrifugal pump creates low pressure to suck water up ✓. If air leaks in, the pump loses its prime ✓ and cannot lift water (or it delivers less and can be damaged) ✓. (3)

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TOTAL: 100

END OF MARKING GUIDELINE